Position Normalization under Proportional Slippage and Partial Invested Fraction

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Abstract

This paper derives a closed-form, piecewise-linear solution for the per-ticker target notional k when rebalancing a portfolio across a changing set of tickers, under proportional slippage costs and an exogenously specified fraction of total wealth to remain invested. Two cached aggregates, β and Φ, are introduced to compress constant and linear-in-k terms so that the root-finding problem reduces to scanning a small number of intervals determined by existing position values. The treatment is algebraically exact, agnostic to implementation language, and accompanied by implementation notes addressing numerical conditioning and unit conventions for slippage.

1 Introduction

A typical rebalancing operation replaces a current set of tickers with a new target set while applying a uniform per-ticker terminal notional k. In the absence of transaction frictions, uniform per-ticker allocation is trivial. When each trade is subject to proportional slippage (a multiplicative factor $1 \pm s_i$), the mapping from pre-trade positions to post-trade positions becomes nonlinear and sign-dependent. Moreover, if a nonzero cash balance exists and only a fraction r of total post-trade portfolio value should be invested in tickers, then cash must be explicitly carried through the algebra. This note provides a compact, publishable derivation of k and a small set of cached aggregates that permit efficient computation.

2 Notation and setup

Let:

- t denote the set of tickers currently held before rebalancing.
- $h \subseteq t$ denote the subset of tickers that will remain (possibly with adjustment) after rebalancing.
- $t \setminus h$ denote tickers in t that are slated for full liquidation.
- b denote the set of new tickers to be purchased (not in t).
- $N := |h| + |b|$ denote the number of tickers that will be held after rebalancing.
- For each ticker i let P_i denote the current dollar value of the existing position (if no existing position then $P_i = 0$).
- For each ticker i let ΔP_i denote the change in position dollar value due to trading, so that the post-trade notional for ticker $i \in h$ equals $P_i + \Delta P_i$.

- For each ticker i let $s_i \geq 0$ denote the proportional slippage rate. Units must be explicit: if s_i is given in basis points then use $s_i \leftarrow s_i/10{,}000$; if given in percent then $s_i \leftarrow s_i/100$. Throughout the algebra s_i denotes the unitless fractional slippage.
- c denotes the cash balance available prior to trading.
- $r \in (0,1)$ denotes the target fraction of total post-trade portfolio value to be invested in tickers.
- k denotes the uniform terminal notional assigned to each ticker in $h \cup b$.

3 Slippage model

Define the directional slippage multiplier

$$S(s_i, \Delta P_i) = \begin{cases} 1 - s_i, & \Delta P_i < 0 \quad \text{(net sell on ticker } i\text{)}, \\ 1 + s_i, & \Delta P_i \geq 0 \quad \text{(net buy on ticker } i\text{)}. \end{cases}$$

Thus selling an intended notional $X > 0$ on ticker i yields cash $(1 - s_i)X$; buying an intended notional $X > 0$ requires paying cash $(1 + s_i)X$.

4 Cash-flow accounting

The cash generated or consumed by modifying existing positions (holds and liquidations) is

$$g_1 = \underbrace{\sum_{i \in h} S(s_i, \Delta P_i)\,(-\Delta P_i)}_{\text{adjustments to retained holdings}} + \underbrace{\sum_{i \in t \setminus h} (1 - s_i)\,P_i}_{\text{proceeds from liquidations}}. \tag{1}$$

New purchases carry notional $Y_i > 0$ (cash outflow). The combined cash change from buys is

$$g_2 := \sum_{i \in b} (-Y_i) = -\sum_{i \in b} Y_i. \tag{2}$$

The starting cash is c; after trading the cash position equals $c + g_1 + g_2$. The total post-trade portfolio value v equals the sum of invested notional plus cash:

$$v = k\,N + \big(c + g_1 + g_2\big). \tag{3}$$

The investment fraction requirement is

$$kN = r\,v = r\big(kN + c + g_1 + g_2\big)$$

provided $N > 0$. If $N = 0$ then $k = 0$ and the entire value is cash.

5 Buy notional substitution

For any new ticker $i \in b$ the buy notional Y_i required to produce a terminal notional k satisfies

$$\frac{Y_i}{1+s_i} = k \quad \Longrightarrow \quad Y_i = k(1+s_i).$$

Summing over b,

$$\sum_{i \in b} Y_i = k\left(|b| + \sum_{i \in b} s_i\right). \tag{4}$$

Therefore (using (2)),

$$g_2 = -k\left(|b| + \sum_{i \in b} s_i\right).$$

6 Per-holding adjustment expression

For $i \in h$ the holding change is $\Delta P_i = k - P_i$. Using the identity

$$S(s_i, \Delta P_i)\,(-\Delta P_i) = \left(1 + \frac{|\Delta P_i|}{\Delta P_i} s_i\right)(-\Delta P_i) = -\Delta P_i - s_i|\Delta P_i|,$$

one obtains, after substituting $\Delta P_i = k - P_i$,

$$S(s_i, \Delta P_i)\,(-\Delta P_i) = -(k - P_i) - s_i|k - P_i|. \tag{5}$$

7 Compressed aggregates β and Φ

Collect constant (i.e. k-independent) terms into a single cached constant β and collect the linear-in-k coefficient into Φ. These cached aggregates permit efficient recomputation:

$$\beta := c + \sum_{i \in t} P_i - \sum_{i \in t \setminus h} s_i P_i, \tag{6}$$

$$\Phi := \frac{N}{r} + \sum_{i \in b} s_i. \tag{7}$$

Interpretation:

- β is the pre-trade scalar that aggregates cash-on-hand c, the current notional across all tickers in t, and the slippage-adjusted reductions from fully liquidated tickers $t \setminus h$.
- Φ is the linear coefficient multipling k that arises from the sum of terminal notional counts and the incremental slippage cost of establishing new tickers.

8 Primary nonlinear equation

Substituting (1), the per-holding expression (5), and the buy substitution into the investment condition and simplifying yields the principal nonlinear equation

$$\boxed{\beta \;=\; \Phi\, k \;+\; \sum_{i \in h} s_i \,|k - P_i|} \tag{8}$$

where, for clarity, the summation on the right-hand side is restricted to the retained holdings h. Equation (8) is the exact balance condition: the left-hand side is a constant with respect to k while the right-hand side is piecewise-linear in k.

Remarks.

- Equation (8) is algebraically equivalent to the investment-fraction form in which r appears explicitly; collecting terms into β and Φ makes the equation concise for implementation.
- The units and conventions for s_i must be consistent across all terms.

9 Piecewise-linear solution

Order the retained holdings' pre-trade notional values

$$P_{(1)} \le P_{(2)} \le \cdots \le P_{(n)}, \qquad n := |h|.$$

The function $k \mapsto \sum_{i \in h} s_i |k - P_i|$ is piecewise-linear, with breakpoints at the $P_{(j)}$. Fix an interval in which the index set of terms with $k \ge P_i$ and $k < P_i$ is constant. Define

$$L := \{i \in h : P_i \le k\}, \qquad R := \{i \in h : P_i > k\},$$

and the cached partial sums

$$S_L := \sum_{i \in L} s_i, \quad S_R := \sum_{i \in R} s_i,$$

$$\Sigma_{P,L} := \sum_{i \in L} s_i P_i, \quad \Sigma_{P,R} := \sum_{i \in R} s_i P_i.$$

On this interval,

$$\sum_{i \in h} s_i |k - P_i| = k(S_L - S_R) + (\Sigma_{P,R} - \Sigma_{P,L}).$$

Substituting into (8) yields a linear equation in k:

$$\beta - (\Sigma_{P,R} - \Sigma_{P,L}) = (\Phi + S_L - S_R)\, k.$$

Provided the denominator is nonzero, the solution on this interval is

$$k = \frac{\beta + \Sigma_{P,L} - \Sigma_{P,R}}{\Phi + S_L - S_R}. \tag{9}$$

The global solution is obtained by scanning the $n+1$ intervals induced by the sorted values $P_{(j)}$ and selecting the expression (9) that lies within the assumed interval bounds. If no interval yields a valid k, the edge intervals (left of $P_{(1)}$ or right of $P_{(n)}$) reduce to the special-case evaluations of (9) with $L = \varnothing$ or $R = \varnothing$ respectively.

10 Special cases and existence

- If $N = 0$ (no tickers to hold) then set $k = 0$, all prior holdings in t are liquidated, and the final cash equals $c + \sum_{i \in t}(1 - s_i)P_i$.
- If $\Phi + S_L - S_R = 0$ on an interval then the right-hand side of (8) is constant on that interval; feasibility requires $\beta = \Sigma_{P,R} - \Sigma_{P,L}$. In practice such exact degeneracies are rare; a numerical guard should treat denominators with a small tolerance.
- Units: if s_i is supplied in basis points the implementer must convert to fractional form by $s_i \leftarrow s_i/10{,}000$.

11 Implementation notes

1. **Cached aggregates.** The constants β and Φ defined in (6)–(7) may be computed in $O(|t|+|b|)$ time; the partial sums $S_L, S_R, \Sigma_{P,L}, \Sigma_{P,R}$ can be accumulated efficiently while scanning the sorted P_i values for $i \in h$.
2. **Numerical tolerances.** Do not test floating-point equality to zero; use tolerances (e.g. 10^{-10} relative) for denominators and interval membership.
3. **Edge-case of r.** If the investment-fraction r is part of the modelling (rather than being folded into β, Φ), ensure $r \in (0, 1)$. As $r \to 1$ the conditioning of intermediate algebra degrades.

12 Conclusion

The present derivation yields a concise, cache-friendly equation $\beta = \Phi k + \sum_{i \in h} s_i |k - P_i|$ whose solution requires only a sorted scan of retained holdings. The separation into β and Φ isolates constant and linear-in-k contributions, enabling efficient recomputation in live trading systems. Practical implementations should include the numerical safeguards noted above.

Appendix: Worked algebraic reduction

The algebraic steps from the investment-fraction condition to (8) are mechanical and are reproduced here for completeness:

$$
\begin{aligned}
kN &= r\,(kN + c + g_1 + g_2) \\
\implies (1-r)kN &= r(c + g_1 + g_2) \\
\implies k &= \frac{r(c + g_1 + g_2)}{(1-r)N} \quad (N > 0, r \in (0,1))
\end{aligned}
$$

Substitute g_1 from (1) and $g_2 = -\sum_{i \in b} k(1 + s_i)$ and perform routine rearrangement collecting k-independent terms into β and linear-in-k terms into Φ. The remaining absolute-value terms originate from the per-holding slippage contributions $-s_i|k - P_i|$, producing (8).